

DAY-6

05-06-2026

Assignment-1

(Q) $Y = A \oplus B \oplus C \oplus D \oplus E \oplus F$

Which of the statements are true?

- (a) Y can be used as odd no. of 1's detector
 (b) Y can be used as even no. of 1's detector
 (c) The number of possible input combinations for $Y=1$ is 32.

(d) The number of possible input combinations for $Y=0$ is 32.

A.

$Y = A \oplus B \oplus C \oplus D \oplus E \oplus F$ [Even no. of inputs (6)]
 $\Rightarrow$ odd no. of 1's detector.

$Y = A \oplus B \oplus C \oplus D \oplus E \oplus F$

For only one "1"

For only Three "1"

For only Five "1"

We get output

(Y) as 1.

$\Rightarrow 6C_1 + 6C_3 + 6C_5 = 6 + 20 + 6 = 32$

Total Comb. $= 2^6 = 64$

$\Rightarrow$ 0's $= 64 - 32 = 32$ [a, c, d].

$$(Q) Y = (A \odot B) + (B \odot C) + (C \odot A)$$

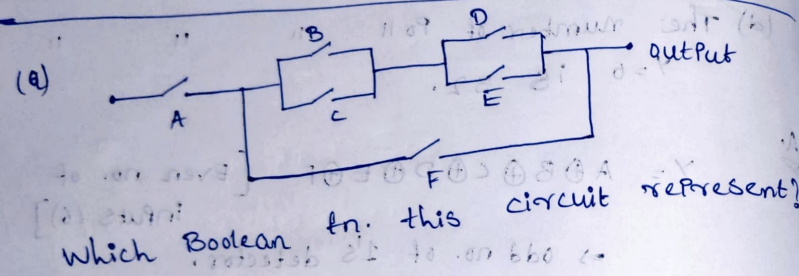
Total no. of NAND gates required to implement Y is?

A. ~~$Y = AB + \bar{A}\bar{B} + BC + \bar{B}\bar{C} + CA + \bar{C}\bar{A}$~~

$$\Rightarrow Y = AB + \bar{A}\bar{B} + BC + \bar{B}\bar{C} + CA + \bar{C}\bar{A}$$

$$= Y + \bar{Y} = 1$$

Zero NAND gates.



A. $A \cdot [(B+C) \cdot (D+E)] + F$

(a) $AB + \bar{A}B + A$ is equivalent to

(a) ~~$A+C$~~ (b) $B+C$ (c) $A+B$ (d) $A+B+C$

A. $\underline{AB + \bar{A}B + A} \Rightarrow AB + C + A$

$$X + \bar{X}Y = X + Y \Rightarrow A[1+B] + C$$

$$\Rightarrow \underline{A+C} \text{ [C]}$$

(a) $(\bar{A}+B)(\bar{A}+C) + \bar{A}B + \bar{A}C$ reduced expression is

(a) $A+B\bar{C}$ (b) $\bar{A}+B\bar{C}$ (c) ~~$\bar{A}+B\bar{C}$~~ (d) $\bar{A}+B\bar{C} + \bar{A}\bar{B}$

A. $\bar{A}\bar{A} + \bar{A}C + \bar{A}B + B\bar{C} + \bar{A}B + \bar{A}C$

$$\Rightarrow \bar{A} + \bar{A}C + \bar{A}B + B\bar{C}$$

$$\Rightarrow \bar{A}[1+C+B] + B\bar{C} \Rightarrow \underline{\bar{A} + B\bar{C} \text{ [C]}}$$

(a) $Y = A[B + \bar{C}(\bar{A}B + \bar{A}C)]$

Reduced expression will have literals—?

A. $Y = A[B + \bar{C}\{(\bar{A}+B) \cdot (\bar{A}+C)\}]$

$$= A[B + \bar{C}(\bar{A}+B\bar{C})]$$

$$= A[B + \bar{C}\bar{A}] = \underline{AB + \bar{A}\bar{C}} \text{ 2 Literals}$$

(a) Boolean expression "Y" will reduce to

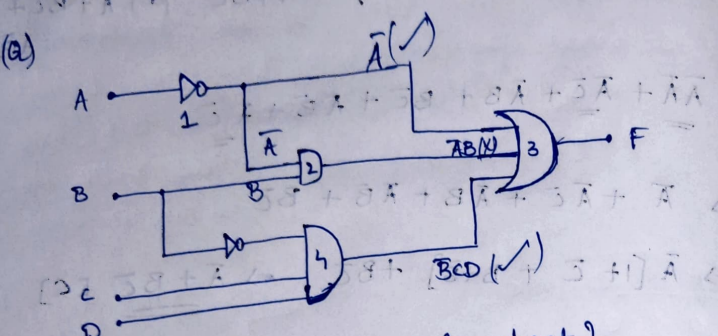
$$Y = \overline{[A+B+C]} + \overline{[\bar{A}+B+C]} + \overline{[A+B+\bar{C}]} + ABC$$

(a) A (b) B (c) C (d) $A+B+C$

A. $Y = \bar{A}B\bar{C} + AB\bar{C} + \bar{A}B\bar{C} + ABC$

$$= \bar{A}B[C+\bar{C}] + AB[C+\bar{C}] = \underline{\bar{A}B + AB}$$

$F = B[A + \bar{A}] = B \cdot 1 = B$



Which gate is redundant?

- (a) 1 (b) 2 (c) 3 (d) 4

A. $F = \bar{A} + \bar{A}B + \bar{B}CD$
 $= \bar{A}(1+B) + \bar{B}CD$
 $= \bar{A} + \bar{B}CD$

In final output, 2 is not coming. So, 2 is redundant.

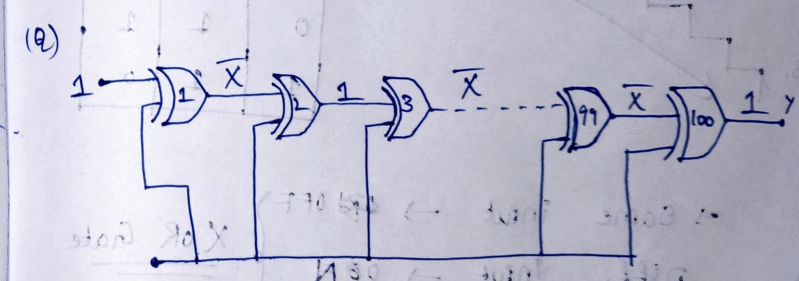
(a) $Y = PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS$
 $PQRS + PQRS$ will be reduced to

- (a) PQ (c) PQRS
 (b) P+Q (d) P+Q+R+S

A. $\Rightarrow Y = PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS$
 $= PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS + PQRS$

$\Rightarrow Y = PQ[RS + RS + \bar{R}S + \bar{R}\bar{S} + \bar{R}ST + RS + \bar{R}ST]$

$\Rightarrow Y = PQ[1 + RST + \bar{R}S + \bar{R}ST] = PQ$



o/p of Y :- (a) X (b) X-bar
 (c) 1 (d) 0

A. odd no. of gates = X
 Even " " " " [100] = 1 [C].

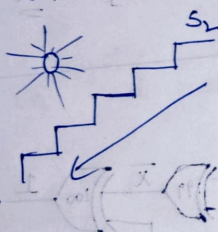
(Q) A bulb in a staircase has 2 switches, one switch being at the ground floor and the other one at first floor. The bulb can be turned ON and also can be turned OFF by ~~one~~ one of switches irrespective of state of other switch. The logic of switching of bulb resembles?

A. (i) going up S_2



S_1	S_2	Y
0	0	0
1	0	1
1	1	0

(ii) Coming down S_2



S_1	S_2	Y
0	0	0
0	1	1
1	1	0

Same input $\rightarrow$ OFF
Diff. input $\rightarrow$ ON

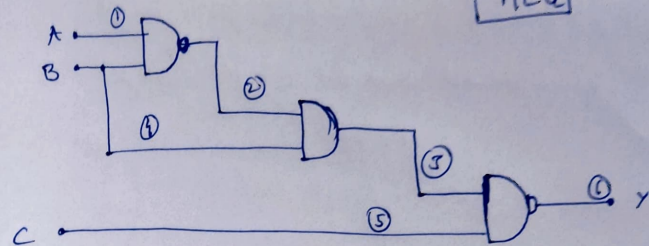
XOR Gate

XOR Gate $\Rightarrow$ Staircase Switch

$\Rightarrow$ Non-equivalence detector

$\Rightarrow$ Odd no. of 1's

(Q)



output "Y" is $\Rightarrow$ (a) $\overline{A} \overline{B} \overline{C}$ (c) $\overline{A} B C$
(b) $\overline{A} + B + C$ (d) $A + \overline{B} + \overline{C}$

A.

1 - A ; 2 $\rightarrow \overline{A \cdot B} = \overline{A} + \overline{B}$

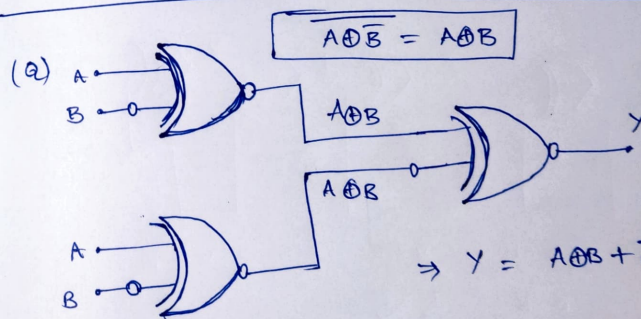
1 $\rightarrow B$; 3 $\rightarrow 2 \cdot 1$

3 $\rightarrow C \rightarrow (\overline{A} + \overline{B}) \cdot B = \overline{A} B$

4 $\rightarrow 3 \cdot 5$

$\rightarrow \overline{A} B C$ [c].

$\rightarrow A + \overline{B} + \overline{C}$ [d].



(a) $A \oplus B$

(b) $A \odot B$

(c) 0

(d) 1

END OF
DAY-6